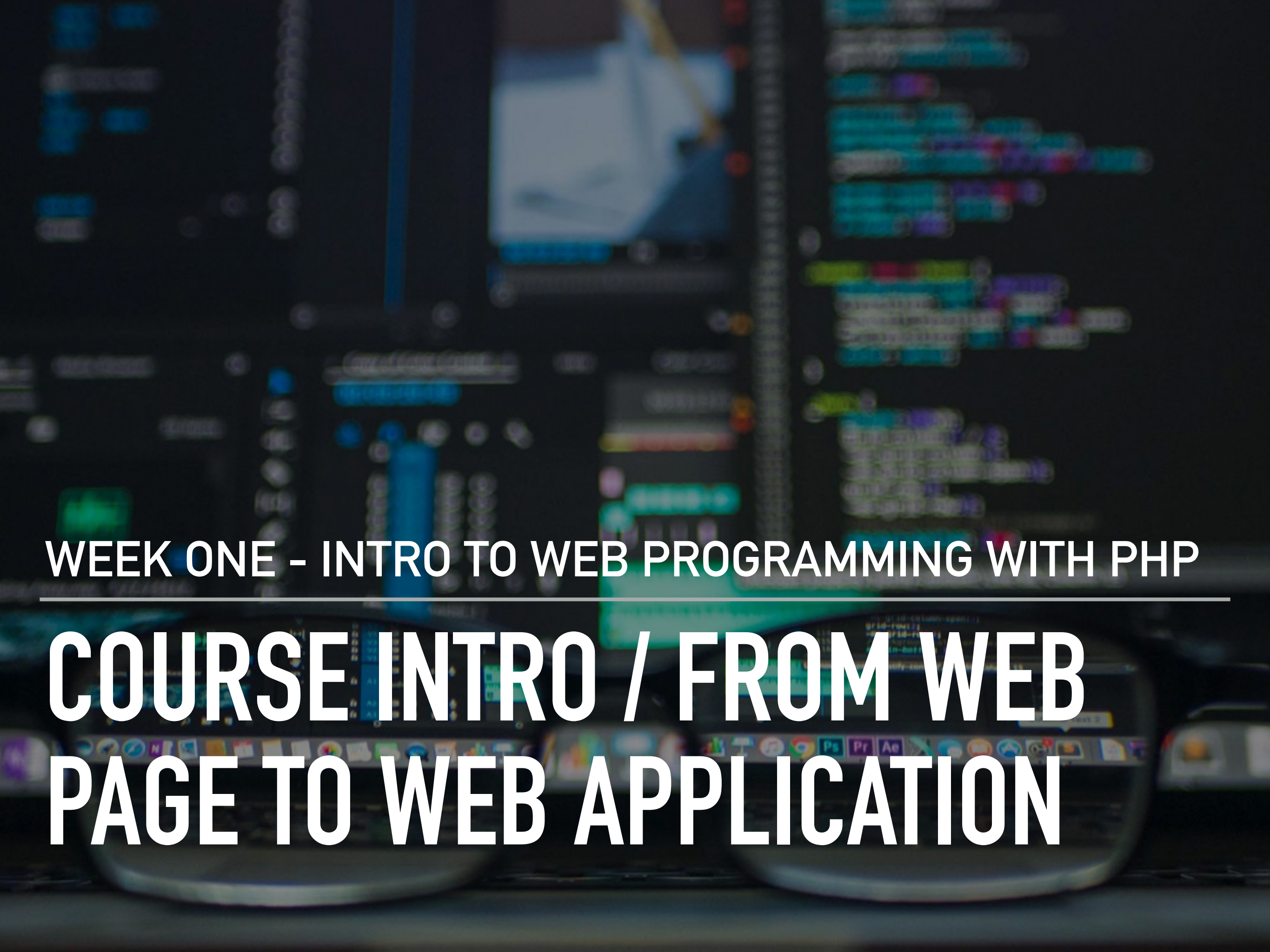


Land acknowledgement

Georgian College acknowledges that all campuses are situated on the traditional land of the Anishnaabeg people. The Anishnaabeg include the Odawa, Ojibwe, and Pottawatomi nations, collectively known as the Three Fires Confederacy. Georgian College is dedicated to honouring Indigenous history and culture and committed to moving forward in the spirit of reconciliation and respect with all First Nation, Métis and Inuit people.



WEEK ONE - INTRO TO WEB PROGRAMMING WITH PHP

COURSE INTRO / FROM WEB PAGE TO WEB APPLICATION



TODAY'S SCHEDULE

1. Welcome/ Course Introduction
2. Syllabus/ Course Outline Review / Blackboard Tour
3. Websites vs Web Applications
4. Static vs. Dynamic
5. Let's Talk PHP
6. Client Side vs. Server Side
7. Setting Up A Local Server
8. Review/Weekly Task List

LEARNING OBJECTIVES – WEEK ONE

- ▶ learn about and set up the tools need for this course (XAMP server, code editor).
- ▶ explore and understand the difference between static and dynamic
- ▶ explore and understand the difference between client-side & server-side scripting



PART ONE – WELCOME!

A LITTLE ABOUT ME

- ▶ Bachelor of Education (Brock University), graduate of the Web Design & Animation (WBAN) program at Georgian (2012 - 2014), currently a enrolled in Master of Education program at Ontario Tech University (I am a student too!)
- ▶ previously worked at Gel Creative, Tyger Shark, 3Sixty Marketing & Element 6
- ▶ teaching at Georgian for 10ish years (CMPG, IMDW & MDEV)
- ▶ cat lady & mom to Felix and Ella
- ▶ coding, coffee, hiking, camping & yoga
- ▶ What I wanted to be when I grew up - a teacher, librarian journalist or graphic designer



INTRODUCE YOURSELF



- ▶ Your name
- ▶ As a child, what did you want to be when you were older?

INTRO TO COMP1006

In this course, you'll build a strong foundation in programming for the web while learning how modern web applications work behind the scenes. Using PHP, you'll learn how the web functions as a development platform and how server-side code makes websites dynamic and interactive. Throughout the course, you'll plan, design, and build web applications that collect user input, work with forms, and store information in databases—developing practical skills used in real-world web development.

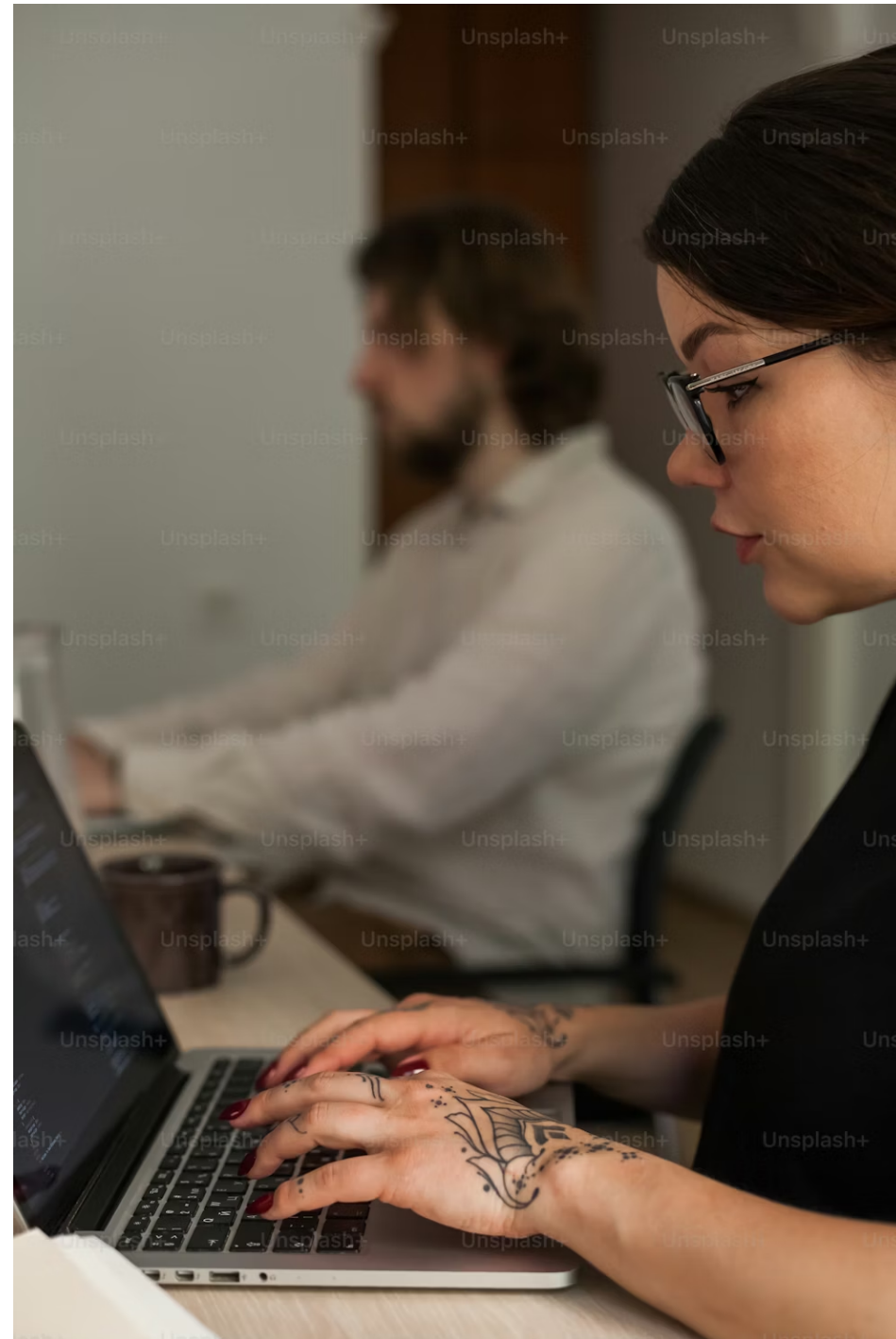


Image Source: [Unsplash](https://unsplash.com/)

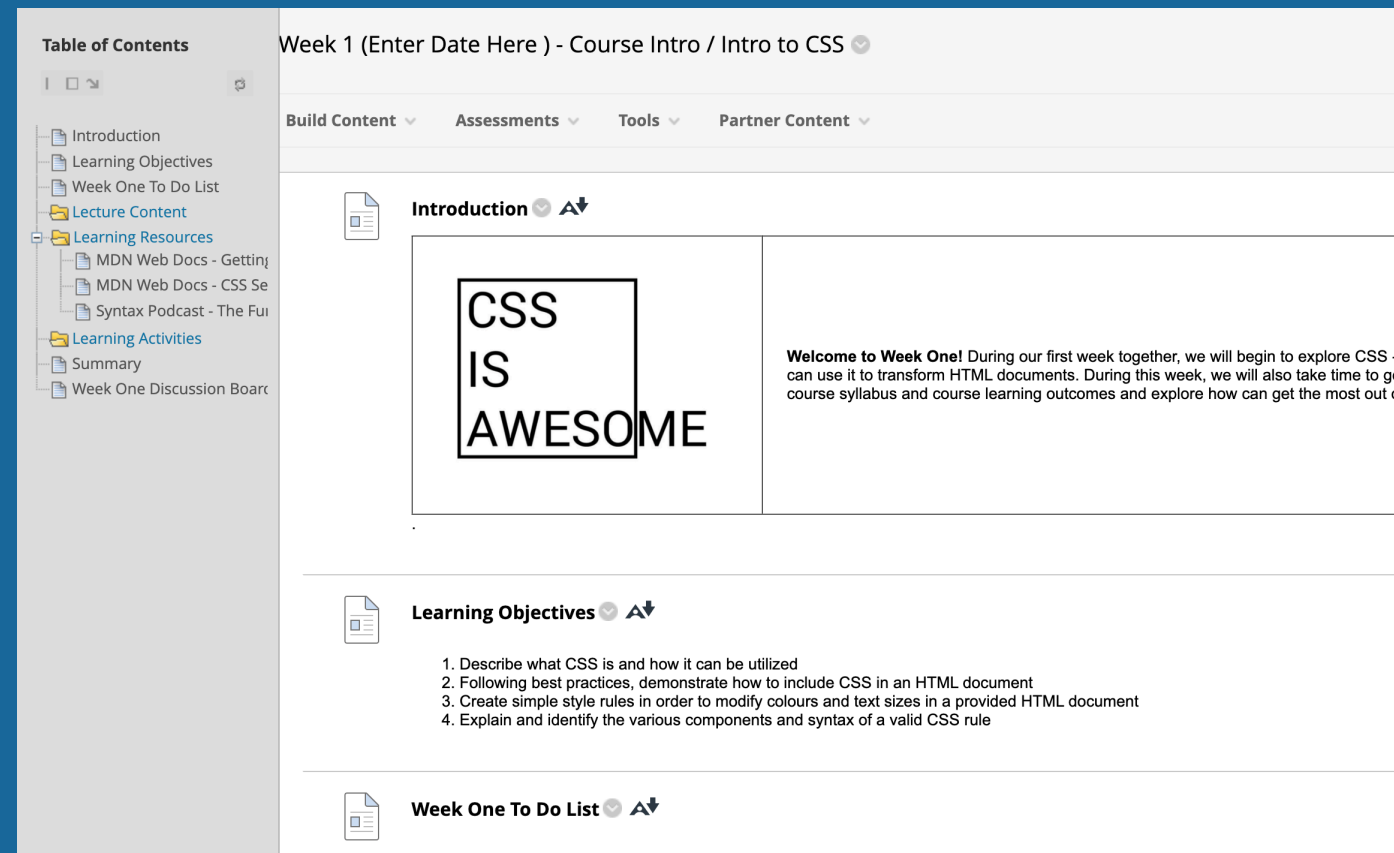
TOPICS WE'LL COVER

- overview of static and dynamic web pages
- server-side scripting with inline code using procedural and object-oriented programming techniques
- programming web forms that connect with web-enabled databases and perform basic CRUD (create, read, update, delete)
- operations
- languages and tools used in server-side programming
- server-side programming resources on the Web
- form input validation on the server-side
- basic security considerations, file permissions, and simple session management

BY THE END OF THIS COURSE, YOU WILL BE ABLE TO:

- A. utilize a variety of basic programming structures (variables, loops, functions etc.) in a popular scripting environment on a Web server;
- B. create scripts that dynamically generate Web pages containing valid HTML (Hypertext Markup Language) clearly separating structure, presentation, and behaviour in the Web browser;
- C. develop scripts that validate form input on the server;
- D. design, program, and implement Web pages that interact with Web-enabled databases performing simple CRUD (Create, Read, Update, Delete) operations;
- E. implement security and file management in a web application;
- F. deploy program to production Web server;
- G. integrate with third party Application Programming Interfaces (API).

BLACKBOARD TOUR, COURSE SYLLABUS & COURSE POLICIES



The screenshot displays a Blackboard course interface for 'Week 1 (Enter Date Here) - Course Intro / Intro to CSS'. On the left, a 'Table of Contents' sidebar lists items: Introduction, Learning Objectives, Week One To Do List, Lecture Content, Learning Resources (MDN Web Docs - Getting, MDN Web Docs - CSS Se, Syntax Podcast - The Fui), Learning Activities, Summary, and Week One Discussion Board. The main content area features three sections: 'Introduction' with a graphic that says 'CSS IS AWESOME' and a welcome message; 'Learning Objectives' with a numbered list of four goals; and 'Week One To Do List'.

Week 1 (Enter Date Here) - Course Intro / Intro to CSS

Build Content ▾ Assessments ▾ Tools ▾ Partner Content ▾

Introduction ✓ A+

CSS IS AWESOME

Welcome to Week One! During our first week together, we will begin to explore CSS. You can use it to transform HTML documents. During this week, we will also take time to go over the course syllabus and course learning outcomes and explore how you can get the most out of this course.

Learning Objectives ✓ A+

1. Describe what CSS is and how it can be utilized
2. Following best practices, demonstrate how to include CSS in an HTML document
3. Create simple style rules in order to modify colours and text sizes in a provided HTML document
4. Explain and identify the various components and syntax of a valid CSS rule

Week One To Do List ✓ A+

A LITTLE BIT ABOUT PBL (PROBLEM BASED LEARNING)

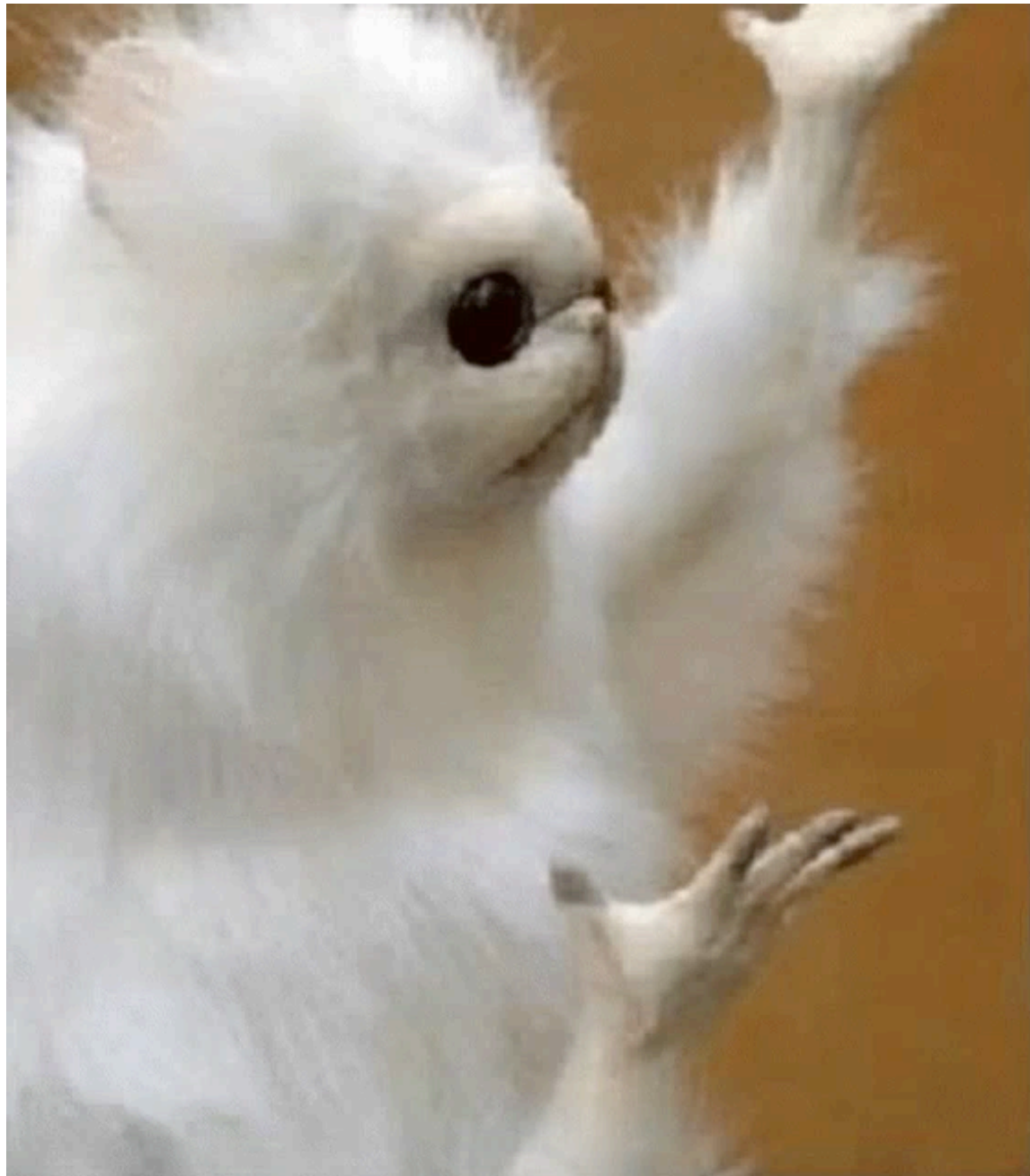


In this course, we will explore a problem-based learning approach inspired by real-world web development. Problem-based learning focuses on learning through working on realistic problems rather than following only step-by-step instructions. You will be encouraged to analyze requirements, research possible solutions, apply new concepts, and refine your work through feedback. This approach helps build critical thinking, problem-solving, and practical skills that are essential in professional web development.

What do you already know?

What do you want to know?

BEFORE WE BEGIN ...



- ▶ What exactly is **web programming**?
- ▶ What are some of the **words** that **come to mind** when you hear this term?
- ▶ What is a **web application**?
- ▶ Are **web applications** and **websites** the same thing?



PART TWO – WEBSITES VS WEB APPLICATIONS

WHERE YOU'RE AT ...



- ▶ Can create awesome, semantic webpages using HTML
- ▶ You use CSS to make those webpages look pretty
- ▶ You include a little JavaScript to make those page do (some) things

HOW CAN WE IMPROVE?

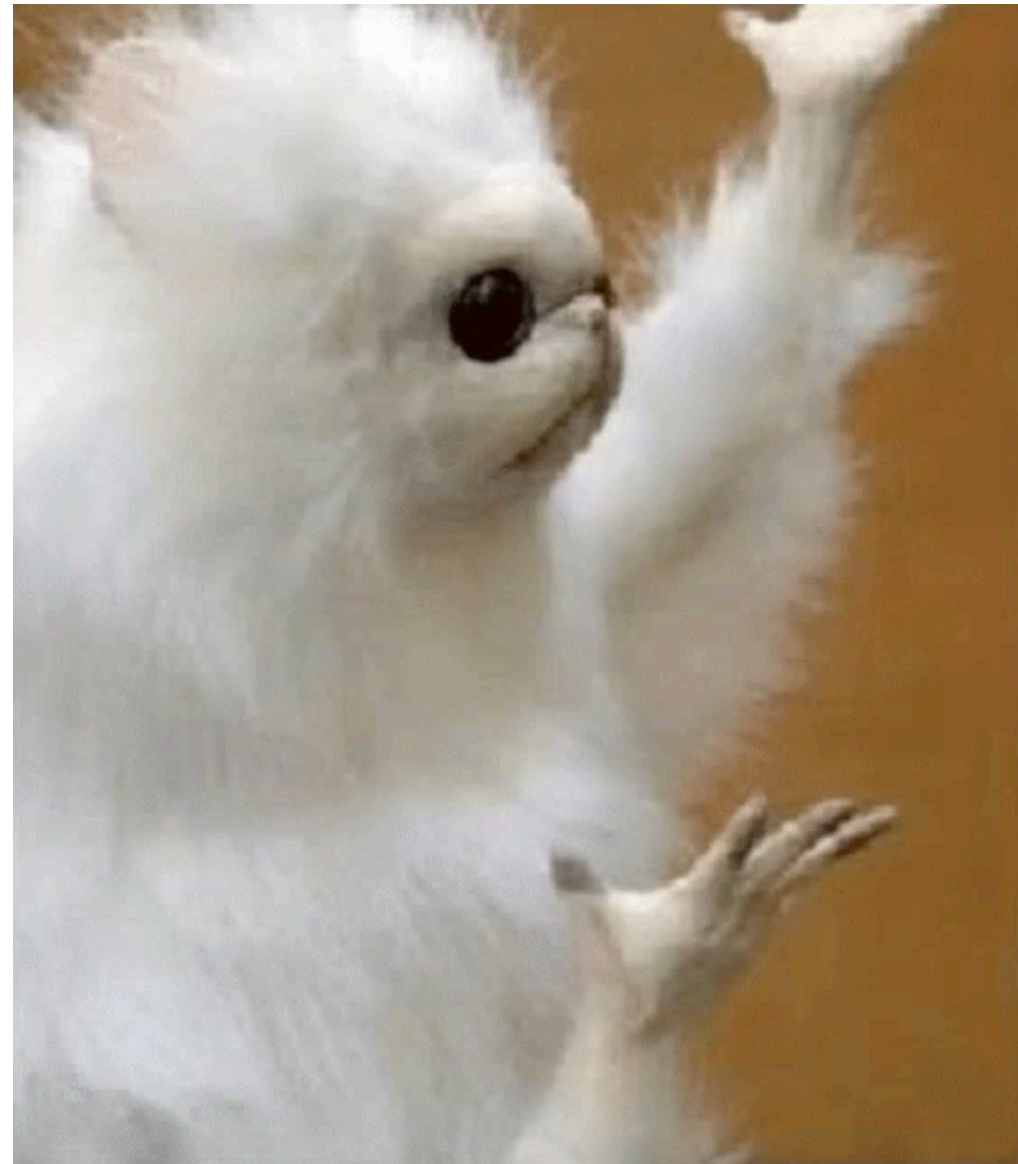


- ▶ Your pages are **static**
- ▶ Users **can't do anything on your page**
- ▶ They can't interact – the **communication is one way**

COMP 1006 - WEEK ONE

FROM BUILDING WEBSITES TO BUILDING WEB APPLICATIONS

WHAT'S THE DIFFERENCE?



<https://youtu.be/ylbQrYhfa18>

STATIC WEB PAGES



Image source: <https://www.pinterest.ca/pin/92464598577679864/>

- ▶ With pure HTML webpages, the **server** simply sends the user static **HTML** that can only display static content
- ▶ Browser requests a page, server responds with HTML

TAKING IT TO THE NEXT LEVEL . . .

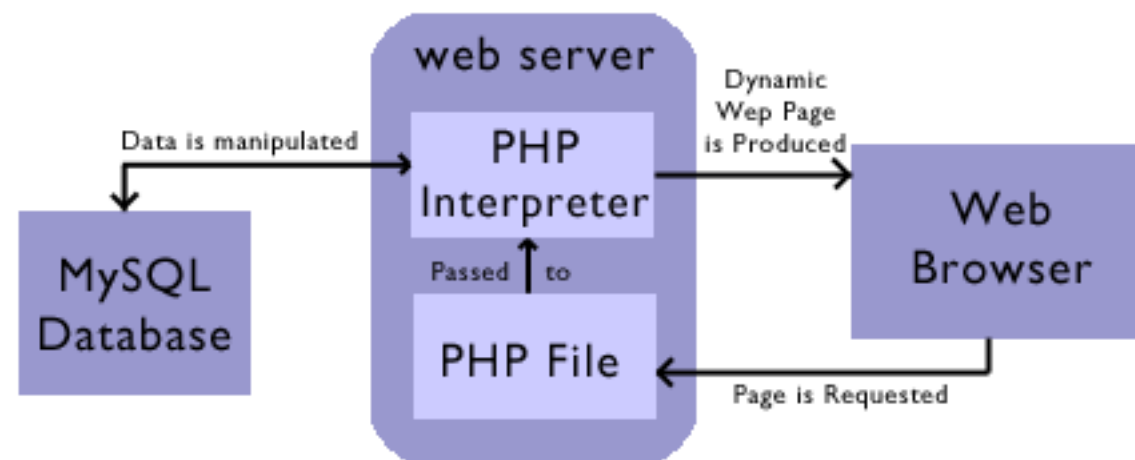


Image source: <https://www.bogotobogo.com/php/php1.php>

- ▶ To turn static pages into **interactive web applications**, the **web server** has to take on a new more dynamic role
- ▶ How? With **PHP**!

WEB APPLICATION ARCHITECTURE



WE ALL USE WEBSITES
& APPLICATIONS . . .

DO WE KNOW

HOW IT ALL REALLY WORKS?

WEB APPLICATION ARCHITECTURE

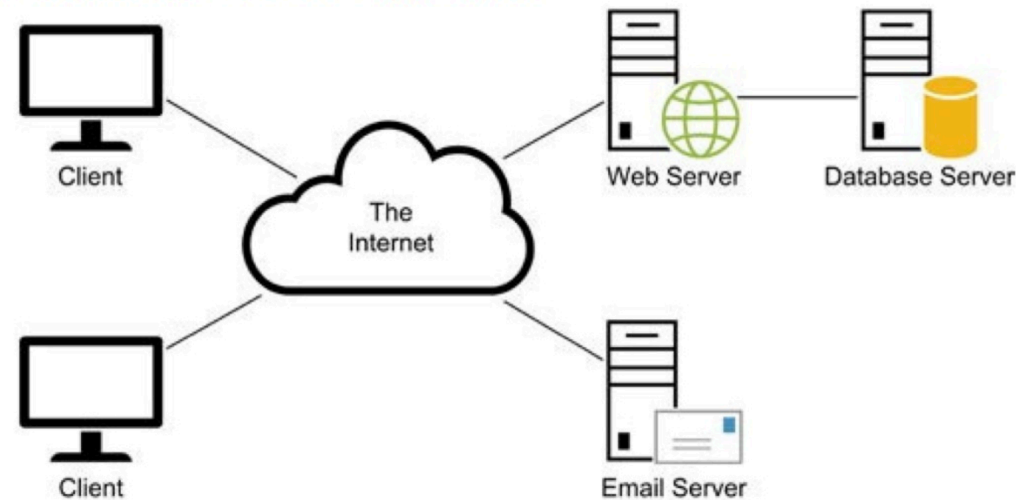
- ▶ Web applications use a client-server architecture.
- ▶ Key terms : server, network, LAN, WAN, ISP, IXP

WEB APPLICATIONS WORK WITH:

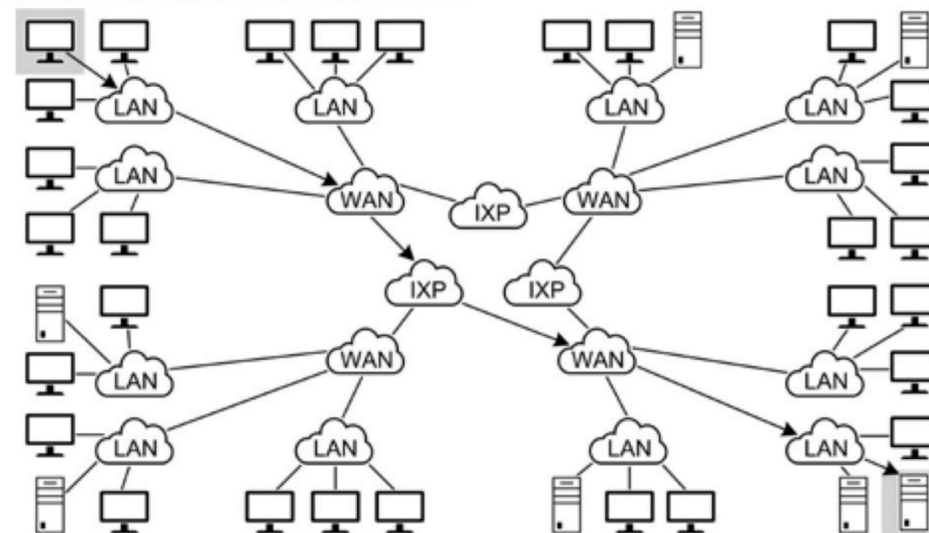
- ▶ **HyperText Transfer Protocol (HTTP)** is the protocol that web browsers and web servers use to communicate. It sets the specifications for HTTP requests and responses.
- ▶ **Transmission Control Protocol/Internet Protocol (TCP/IP)** is a suite of protocols that let two computers communicate over a network

WEB APPLICATION ARCHITECTURE

The architecture of a web application

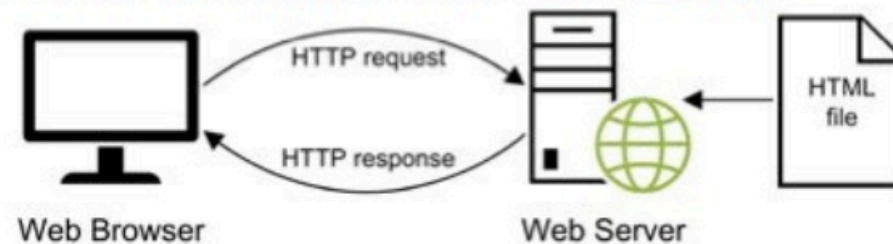


The architecture of the Internet



HOW **STATIC** PAGES ARE PROCESSED

How a web server processes a static web page



A simple HTTP request

```
GET / HTTP/1.1
Host: www.example.com
```

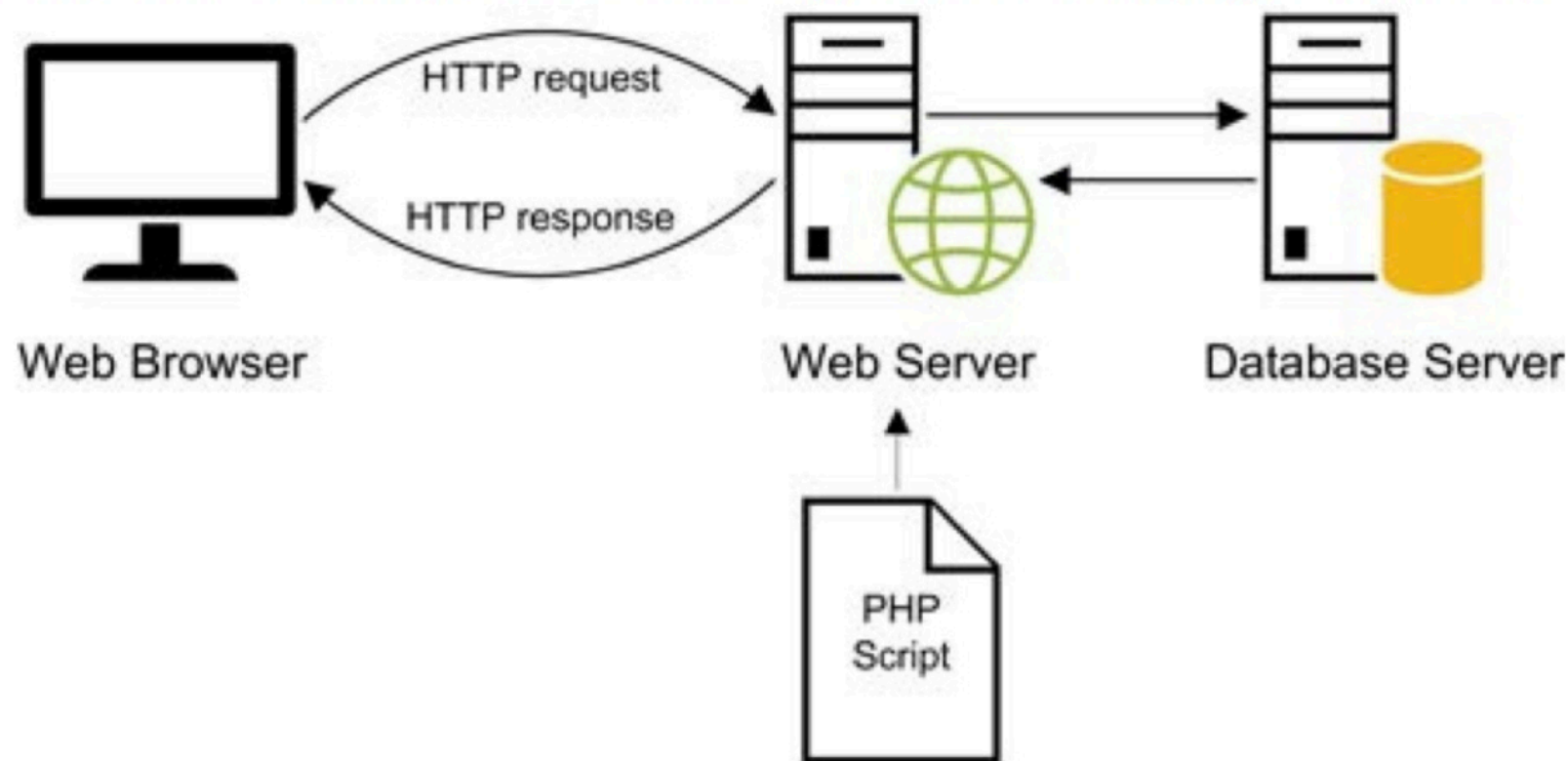
A simple HTTP response

```
HTTP/1.1 200 OK
Content-Type: text/html
Content-Length: 136
Server: Apache/2.2.3

<html>
<head>
  <title>Example Web Page</title>
</head>
<body>
  <p>This is a sample web page</p>
</body>
</html>
```

HOW **DYNAMIC** PAGES ARE PROCESSED

How a web server processes a dynamic web page with PHP





PART THREE- GETTING TO KNOW PHP



ACTIVITY ONE: WHAT CAN WE LEARN ABOUT PHP IN 10 MINS?

A LITTLE ABOUT PHP

- ▶ PHP (which - fun fact -stands for hypertext preprocessor) script runs on the server and **can alter or generate HTML code**
- ▶ PHP scripts are **stored on the server where they are processed** and then delivered to the browser as HTML
- ▶ Contain **both HTML code and PHP script** that determines how the HTML code is manipulated

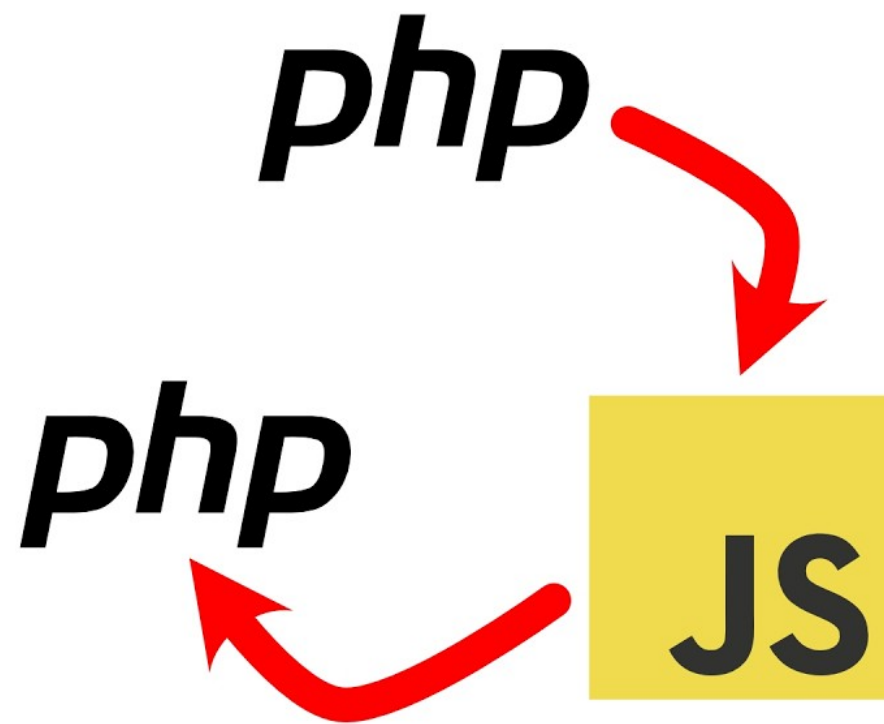
A LITTLE ABOUT PHP

- ▶ PHP can store and retrieve data from a database and incorporate the data into HTML code that it generates
- ▶ Dynamic HTML pages change in response to programmatic logic in PHP scripts, making them incredibly flexible

A LITTLE ABOUT PHP

- ▶ PHP code is always enclosed by `<?php and ?>`
- ▶ Every PHP statement must `end with a semicolon`
- ▶ If there is any PHP code in a webpage, it's good practice to name the file on the webserver with the `.php extension instead of .html`
- ▶ PHP `variable names must begin with a dollar sign $`

BUT WAIT... WHY PHP?

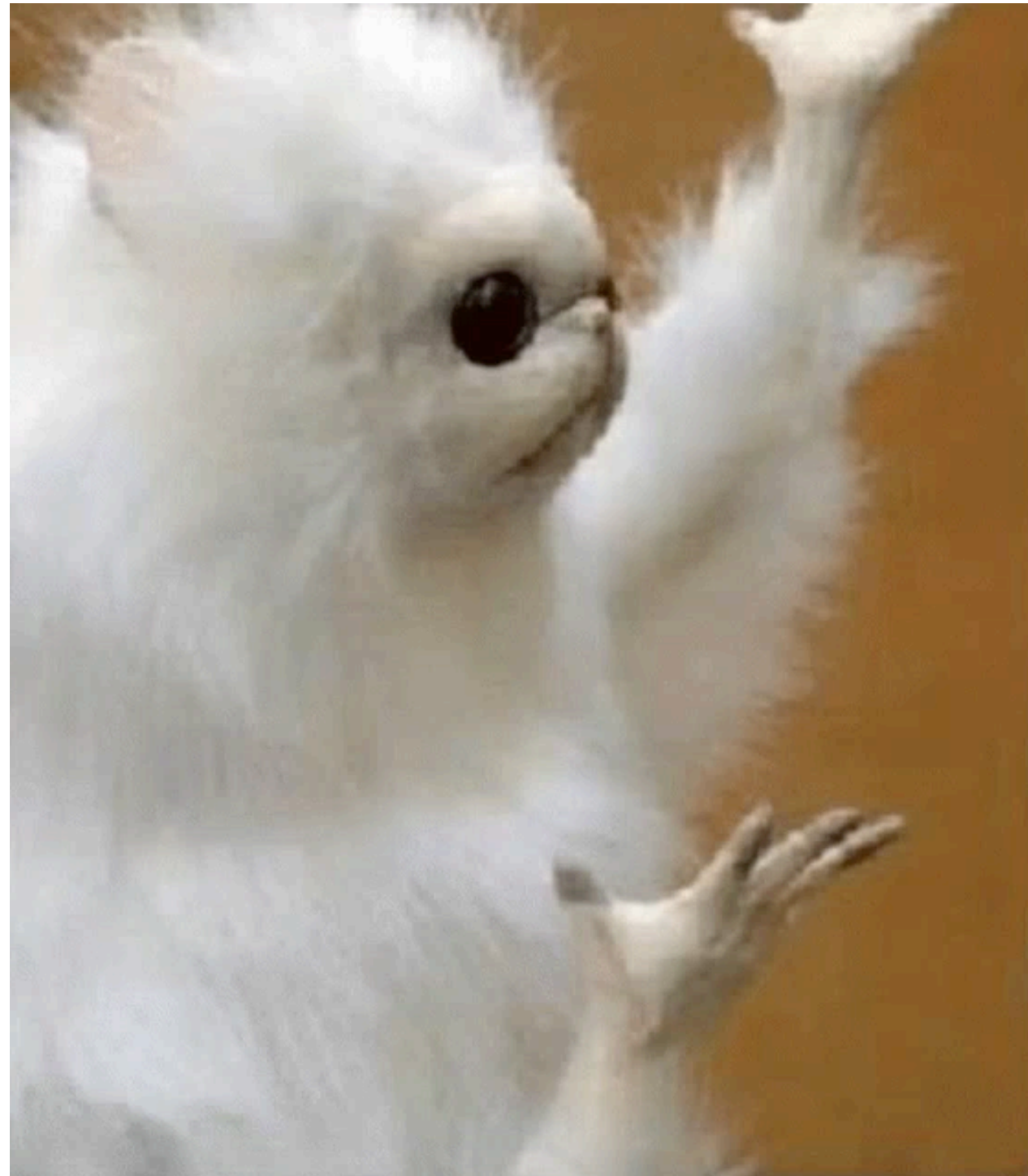




READY TO TRY SOME **PHP**?

**LET'S TAKE A LOOK
IN THE BROWSER.**

WHA HAPPENED?



HTML ACTS ON THE
CLIENT

PHP ACTS ON THE
SERVER

RUNNING YOUR PHP SCRIPTS

- ▶ in order for PHP to run, we need some kind of web server with PHP support
- ▶ we'll be using a local server (WAMP/XAMP/MAMP)
- ▶ we'll also be using AWS cloud hosting

SET UP YOUR LOCAL SERVER

Blackboard > Week One > Additional Resources
> Mac Setup

OR

Blackboard > Week One > Additional Resources
> Windows Set Up



PART FOUR – SUMMARY, TO DO & NEXT WEEK

WHAT YOU LEARNED

1. Course Intro & Getting To Know Each Other
2. Websites vs. Web Applications
3. Getting to Know PHP

EXTEND YOUR UNDERSTANDING

Weekly Learning > Week One > [Additional Resources](#)

BEFORE OUR NEXT CLASS, PLEASE

1. Complete the Week One exit ticket
2. For a deeper dive, check out additional resources
3. Add any outstanding questions, ideas or cool finds to the Week One Discussion Board

NEXT WEEK >

IN PERSON

We'll cover:

- ▶ PHP Basics & Object Oriented Programming